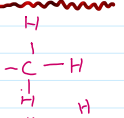
February 16, 2026 5:12 PM

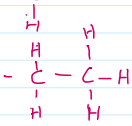
1) Alkanes

- ↳ All carbons are bonded to other atoms by single bonds
- ↳ This means that each carbon atom has the maximum of hydrogen atoms bonded to it.
- ↳ No more hydrogen atoms can be added
- ↳ Thus these are called saturated hydrocarbons.

- ↳ Prefix → # of C atoms in the chain
- ↳ Suffix → -ane

Alkane formula ⇒ $C_n H_{2n+2}$

Methane → C_1H_4 ⇒ 

Ethane → C_2H_6 ⇒ 

Propane → C_3H_8

Butane → C_4H_{10}

Pentane → C_5H_{12}

Hexane → C_6H_{14}

Heptane → C_7H_{16}

Octane → C_8H_{18}

Nonane → C_9H_{20}

Decane → $C_{10}H_{22}$

Methyl-	$-CH_3$
Ethyl-	$-CH_2-CH_3$
Propyl-	$-CH_2-CH_2-CH_3$
Butyl-	$-CH_2-CH_2-CH_2-CH_3$
Pentyl-	$-CH_2-CH_2-CH_2-CH_3$
Hexyl-	$-CH_2-CH_2-CH_2-CH_2-CH_3$

2) Alkenes

- ↳ Hydrocarbons that contain one or more C=C
- ↳ C_nH_{2n} n → no. of Carbon atoms

3) Alkynes.

- ↳ Hydrocarbons that contain 1 or more C≡C
- ↳ C_nH_{2n-2}

Naming.

↳ 1) Parent chain must be alkene or alkyne

2) When numbering start with the end closest to the multiple bond [C=C or C≡C].

3) Multiple bonds take priority over branches [methyl, ethyl...]

4) If there are multiple double or triple bonds, indicate it with the prefixes, di-, tri-, etc.